

How we judge whether our model is good

The benchmarking problem in long-short equity, what metrics we use, and how we test against the time-window cherry-picking critique.

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The short answer

A long-short equity strategy is a fundamentally different investment than the S&P 500.

Comparing their raw returns is the wrong question. We evaluate at three levels: **(1) does the model predict returns better than a coin flip, (2) does the strategy make money on a risk-adjusted basis, and (3) is the result robust to which 5-year window we picked.**

Bootstrap CIs, deflated Sharpe, and Fama-French regressions answer each question in turn.

Our canonical model (Phase 8) passes (1) and (2) cleanly on the 10-year window; the more stringent factor-adjusted test for (2) is borderline.

1. Why we can't just compare to the S&P; 500's return

1.1 What a long-short portfolio actually is

Our strategy holds equal-dollar amounts long and short. If you put \$1,000 into the strategy:

- \$1,000 of cash sits in your account
- You borrow \$1,000 of stocks and sell them (the short leg)
- You then own \$1,000 of cash + \$1,000 of bought stocks (long leg) - \$1,000 of stocks owed (short leg). Net asset value = \$1,000 of cash.
- The strategy's P/L over the month is (long-leg return) - (short-leg return), expressed as a percentage of NAV. The market's overall direction barely matters by construction.

If S&P 500 rises 10% in a month and our long leg is up 12% while our short leg is up 9%:

Long P/L = $+12\% \times \$1000 = +\120

Short P/L = $-9\% \times \$1000 = -\90 (we LOSE when shorted stocks rise)

Net P/L = $+\$30 = +3\%$ on \$1,000 NAV.

If S&P 500 falls 10% and long is down 12%, short is down 9%:

Long P/L = $-12\% \times \$1000 = -\120

Short P/L = $+9\% \times \$1000 = +\90 (we GAIN when shorted stocks fall)

Net P/L = $-\$30 = -3\%$ on NAV.

Symmetric. The strategy doesn't care about market direction - it cares about the SPREAD between winners and losers.

1.2 Why comparing to S&P; 500 returns is the wrong question

The S&P 500 in 2019-2024 returned ~13% per year. Our long-short strategy returned ~5%. A naive reader would say "the strategy underperformed." That's wrong, for two reasons:

- **The strategies take different amounts of risk.** The S&P 500 has annualised volatility ~18% and a max drawdown of ~34% in 2020. Our strategy has volatility ~9% and max

drawdown ~9%. We took roughly half the risk and made roughly a third of the return. What matters is return PER UNIT of risk.

- **The strategies have different exposures.** The S&P 500 is a pure long bet on the US market. Our strategy is approximately market-neutral (net beta of +0.05 to +0.10). They're not comparable as investments any more than a 30-year bond is comparable to a venture capital fund.

The correct comparison is risk-adjusted. An investor could lever our +0.66 Sharpe strategy 2x to roughly match the S&P 500's annualised return -- with comparable volatility but a much shorter maximum drawdown. Or they could blend our market-neutral return with their existing portfolio to add a non-correlated return stream. Neither of those uses the S&P 500's raw return as the benchmark.

2. The three levels of evaluation we use (Framework section 8.2)

The Project Framework prescribes a 3-level evaluation, each answering a different question.

2.1 Level 1: Did the model predict returns better than chance?

Question: ignoring the portfolio entirely, can our XGBoost model rank stocks better than a random number generator?

Two metrics:

- **Out-of-sample R^2 (Gu-Kelly-Xiu version):** $1 - \text{SUM}((y - \text{pred})^2) / \text{SUM}(y^2)$. Numerator is squared forecast error; denominator is total squared return. $R^2 > 0$ means we beat predicting zero. For individual stock returns, GKX (2020) reports OOS R^2 in the 0.1-0.6% range across all their models. Ours: Phase 8 = -2.0% (negative because our tree model makes large predictions; squared error punishes that even when ranking is good).
- **Information Coefficient (IC):** per-month rank correlation between predicted scores and realised returns, averaged across months. $IC > 0$ means we rank stocks correctly on average. Production cross-sectional models typically score 0.01-0.03. Ours: Phase 8 = +0.0123. Small but positive.

2.2 Level 2: Did the strategy generate alpha?

Question: when we actually trade on the predictions with realistic costs, does the portfolio make risk-adjusted money?

Primary metric: Sharpe ratio. (Annualised return / annualised volatility). Captures return per unit of risk. A Sharpe of 1.0 is considered very good for a long-short equity strategy; 0.5-0.8 is solid. Ours: **+0.66** on the canonical 2019-2024 test window, **+0.79** on the 2015-2024 long-OOS window.

Secondary metrics:

- **Maximum drawdown:** worst peak-to-trough loss. Lower is better. S&P 500's 2019-2024 max drawdown was -34%; ours is -9%.
- **Calmar ratio:** annualised return / |max drawdown|. Another risk-adjusted measure that punishes deep drawdowns more harshly than Sharpe does.
- **Average monthly turnover:** measures transaction cost drag. Ours: ~1.77 (i.e. we replace ~88% of positions each month). High but absorbed by the 10 bps cost assumption.

2.3 Level 3: Does the regime overlay add value?

Question: do market-condition-aware leverage adjustments improve the strategy beyond running it at constant leverage?

Not yet evaluated in this project because Person C's GMM regime model isn't wired through Bowen's backtest engine yet (the `regime_fn` parameter exists but the current canonical run uses it as `None`). Once Layer 3 is wired, we compare Sharpe with and without the regime overlay on the same predictions.

3. The time-window critique and how we handle it

3.1 The COVID problem (and 2022 and 2020 banking crisis)

The framework says "test window 2019-2024." That window contains three unique market events:

- **Q1 2020 COVID crash:** S&P 500 fell ~34% in 5 weeks. Volatility spiked to crisis levels. Cross-sectional patterns broke down (everything sold off together).
- **2022 bear market:** the largest tech selloff since the dot-com era. Growth dramatically underperformed value -- the opposite of the 2015-2019 regime.
- **March 2023 banking crisis:** SVB / Signature / First Republic collapsed. Financial sector specific.

If our strategy works in this 5-year window, a skeptic could argue we got lucky -- maybe our model happens to do well during these specific events but would fail in a different regime. We address this two ways:

3.2 Robustness check #1: longer out-of-sample window

Even within the data we have (2005-2024), we report metrics on **both** the framework's 5-year window AND a 10-year window:

Window	Years	n_month	Includes	Sharpe (ca)
2019-2024 (framework test)	5	60	COVID + 2022 + SVB	+0.66
2015-2024 (long-OOS)	10	120	2015 China selloff + 2016 election + 2017 COVID + 2022 + SVB	+0.79

Both windows give positive Sharpe. The long-OOS Sharpe (+0.79) is actually **HIGHER** than the short-window one (+0.66) -- the model performed even better on 2015-2018 than on 2019-2024. So the framework's 5-year test window is, if anything, a slightly hostile subset.

3.3 Robustness check #2: block bootstrap confidence intervals

The framework section 8.3 explicitly prescribes block bootstrap. We take the 120 monthly returns of the long-OOS window, resample them in 6-month blocks (preserving short-horizon autocorrelation), and recompute the Sharpe 10,000 times. The 5-95% percentile of those 10,000 Sharpes is the bootstrap CI:

Window	Observed Sharpe	5-95% Bootstrap CI	P(SR ≤ 0)
2019-2024	+0.69	[+0.05, +1.09]	3.7%
2015-2024	+0.79	[+0.36, +1.13]	0.14%

On both windows the CI excludes zero -- the strategy is statistically distinguishable from random. On the 10-year window, the probability that the true Sharpe is ≤ 0 is 0.14% -- a very strong rejection.

3.4 Robustness check #3: deflated Sharpe (multi-test correction)

Bailey & López de Prado (2014) point out that we tried multiple model configurations along the way (Phase 1 → 1.5 → 2 → 3b → 3c → 8). Each variant is a hypothesis test. The MAXIMUM Sharpe across 6 attempts is biased upward -- some of it is just luck of picking the best of 6 noisy estimates.

The deflated Sharpe corrects for this:

- Compute expected maximum Sharpe given N=6 random configurations with the observed Sharpe-spread variance (Bonferroni-style)
- Subtract that expected-max from the observed Sharpe
- Penalise further for non-normal moments (skewness, kurtosis) of the return series
- Convert to a probability via the standard normal CDF -- the DSR.

DSR > 0.95 means the strategy is significant at 5% AFTER adjusting for variant search:

Window	DSR (Phase 8)	Threshold	Verdict
2019-2024	0.85	0.95	Borderline (not significant)
2015-2024	0.96	0.95	Significant ✓

The 10-year DSR clears the threshold. The 5-year window is borderline but the bootstrap test on that window separately shows $P(SR \leq 0) < 5\%$.

4. The hardest test: is the Sharpe genuine skill, or factor exposure?

4.1 The Fama-French question

A long-short strategy can produce a positive Sharpe in three ways:

- **Genuine cross-sectional skill** -- the model picks winners and losers based on real, repeatable patterns.
- **Factor-tilt accidents** -- the strategy happens to short value stocks and long growth stocks during a period when growth dominated value. Looks like skill in-sample but is really captured by Fama-French factor premia.
- **Survivorship / data bias** -- the strategy worked because we didn't properly handle delisted stocks, banking crises, etc.

We control for (3) by construction: CRSP price data includes delisted stocks with their bankruptcy returns; Sharadar SF1 covers delisted fundamentals; the fja05680 membership table is point-in-time. We test for (2) by regressing strategy returns on Fama-French factors:

4.2 The regression

For each monthly portfolio return $r_{p,t}$ in the test window, we run:

$$r_{p,t} - r_{f,t} = \alpha + \beta_{\text{Mkt-RF}}(\text{Mkt-RF})_t + \beta_{\text{SMB}}(\text{SMB})_t + \beta_{\text{HML}}(\text{HML})_t + \varepsilon_t \text{ (FF3)}$$

or, adding profitability (RMW) and investment (CMA):

$$r_{p,t} - r_{f,t} = \alpha + \beta_{\text{Mkt-RF}}(\text{Mkt-RF})_t + \beta_{\text{SMB}}(\text{SMB})_t + \beta_{\text{HML}}(\text{HML})_t + \beta_{\text{RMW}}(\text{RMW})_t + \beta_{\text{CMA}}(\text{CMA})_t + \varepsilon_t \text{ (FF5)}$$

The factor returns come from Ken French's data library (free, monthly, since 1926). Newey-West HAC standard errors handle autocorrelation. We focus on:

- **α (alpha):** the part of the strategy's return NOT explained by factor exposures. If α is significantly positive, we have skill beyond factor replication.
- **β coefficients on each factor:** tells us WHICH factors the strategy is effectively replicating.
- **R^2 of the regression:** how much of our return variance is explained by factors.

4.3 The results — the most consequential numbers in the project

	FF3 (test)	FF3 (long-OOS)	FF5 (test)	FF5 (long-OOS)
α (annualised)	+1.95%	+2.88%	+1.72%	+2.68%
α t-stat	+0.46	+1.17	+0.44	+1.14
α p-value	0.65	0.24	0.66	0.25
Mkt-RF β	+0.11	+0.10	+0.16 ***	+0.14 ***
SMB β	0.00	+0.07	-0.03	+0.06
HML β	-0.17 ***	-0.13 **	-0.29 ***	-0.26 ***
RMW β			-0.12	-0.05
CMA β			+0.34	+0.30 *

R ²	0.12	0.10	0.19	0.16
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The honest story: alpha is positive (+1.7 to +2.9% per year) but NOT statistically significant after factor adjustment (t-stat between 0.4 and 1.2 on all four specifications). The dominant factor exposure is the value factor: HML loading of -0.17 to -0.29 with t-stats of -2.7 to -4.6. The strategy is persistently **short value**, and value underperformed growth massively in 2015-2024.

Interpretation: the ML model has empirically discovered the growth-over-value premium of this period, plus a smaller residual skill component. A passive HML-short ETF would have captured most of the same Sharpe at much lower complexity.

This is the kind of finding methodologically-careful financial-ML papers routinely report. It's not a failure -- it's a correctly-diagnosed limitation. Future work (Bowen's Layer 3 sector-neutral construction, or an explicit factor-hedge) could push alpha higher by removing the factor tilts.

5. The honest scorecard

Compact summary of every claim we make, with the evidence behind it:

Claim	Test	Result	Verdict
The model predicts better than random	Can random	+0.0123	weak but positive ✓
The strategy makes money (5yr)	Sharpe (5yr)	+0.66	decent ✓
Holds on a longer window	Sharpe (10yr)	+0.79	better, not worse ✓
Statistically distinguishable from 0	Bootstrap CI excludes 0	[0.36, +1.13]	$P(SR \leq 0) = 0.14\%$ ✓
Survives multi-test correction	Deflated Sharpe > 0.95 (5yr)	DSR = 0.96	passes ✓
Same on 5yr window	DSR > 0.95 (5yr)	DSR = 0.85	borderline ✗
Survives factor adjustment	FF5 alpha significant	t = 1.14, ns	fails ✗
Drawdown smaller than market	max DD	-9%	vs S&P 500's -34% ✓
Empirically market-neutral to S&P 500	β to S&P 500	+0.09 (t=1.99)	small, borderline ✓

Six green checks, one borderline, two open issues. The two open issues are the FF5 alpha (not significant after factor adjustment, suggesting much of the Sharpe is replicable by a HML-short) and the 5-year-window DSR (borderline). Both are honest limitations to disclose in the final report, not bugs.